Statement of Work – *MEAD ARIA and Field Test Logistics*

March 2025

Subgrant under **MEAD: Mars Exploring Analog Drilling**, awarded 11/24 under Solicitation Number NNH23ZDA001N-PSTAR

Effective date: 1 March 2025
End date: 31 October 2027

No.	Task	FY25	FY26	FY27
1	ARIA – MEAD Integration (Base Period): (a) Planning and coordination with MEAD team and PI regarding 8/25 field test (b) Develop a plan with the MEAD team and PI to be followed for any necessary field sample transport and permitting (c) Support field readiness review (6/25) and periodic technical consultations. (d) Travel to/from Haughton Crater with ARIA instrument and any necessary support equipment (8/25) (e) Demonstrate (standalone) ARIA (both standoff and iCERS modes) under Mars-analog field operating conditions. (f) Evaluate impact crater target materials determining mineral compositions and trace organics and volatiles, including organic thermal maturity, across frozen boundary.	**3/1/25 – 10/31/25**		
2	MEAD Year 1 Field Logistics Support (Option 1): (a) Coordinate with the MEAD PI to insure all required field research licenses and permits are acquired and presented (b) Provide MEAD logistic support including purchase of supplies and materials for the research field camp.			
2	ARIA Data Studies and Maintenance (Option 2): (a) Curation of field samples and of ARIA datasets (as per MEAD OSDMP) (b) ARIA post-field maintenance and upgrades as necessary (c) Study Y1 field data for evidence of any variation of thermal maturity and biosignatures across the active layer		**11/1/25 – 10/31/26**	

	(d) Present pertinent Y1 results at the Astrobiology Science Conference (5/26) (e) Lab validation between ARIA and the existing SWRI HORIBA Raman system, using Y1 field samples (7/26)			
3	ARIA Field Testbed Integration (Option 3): (a) Planning and coordination with MEAD team and PI regarding lander mockup interfaces and mounting for 8/27 field test (b) As-needed revise the field sampling plan with the MEAD team and PI (c) Support field readiness review (6/27) and periodic technical consultations. (d) Travel to/from Haughton Crater with ARIA instrument and any necessary support equipment (8/27) (e) Demonstrate ARIA - Lander mockup integration and operations in a relevant Mars analog field environment with the integrated MEAD automated drill and robotic sample transfer (f) Evaluate impact crater target materials determining mineral compositions and trace organics and volatiles, including organic thermal maturity, across frozen boundary. (g) Curation of field samples and of ARIA datasets (as per MEAD OSDMP)			**11/1/26 – 10/31/27**